

STAGE 3 OF THE DESIGN SEQUENCE · V1.2

Business Rules & Domain Invariants

The enforceable constitution of the domain

Stage 2 defined what exists. This stage defines what is allowed — every policy, invariant, state transition, transaction boundary, and authorization rule written so it can be enforced by code rather than by convention. Updated with the JKUAT decisions of 30 July 2026.

Product	MentorMAMA — Digital Mentorship Toolkit for Safer Midwifery Clinical Placement
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Document type	Stage 3 of the design sequence: Business Rules & Domain Invariants
Precedes	Stage 3.5 (Use Cases & Sequence Diagrams) → Stage 4 (Module Architecture) → Stage 5 (Database Architecture)
Builds on	Stage 1 Business Discovery Report v1.0; Stage 2 Domain Model (Parts 1 & 2); Developer Concept Note v1.0; SADD v1.0
Version	1.2 (incorporates JKUAT decisions of 30 July 2026 — see Appendix B)
Date	July 2026

1. Purpose of This Stage

Stage 2 defined **what exists**. This stage defines **what is allowed**.

Every rule in this document is written to be enforced by code, not by convention. Each one carries an identifier, an owner (the layer that enforces it), and a failure mode (the domain error raised when it is violated). A rule that cannot be stated with those three properties is not a rule — it is a preference, and it belongs in the product backlog, not here.

This matters more for MentorMAMA than for an average workflow product for two reasons carried forward from Stage 1:

1. **Two independent institutions share one object.** A university and a hospital both act on the same Placement while having zero authority inside each other's jurisdiction. Rules are the only thing preventing one side from silently overriding the other.
2. **Part of the data is safeguarding-sensitive.** Escalations and student feedback carry real-world consequences for real people. "The UI does not show that button" is not an access-control model.

1.1 How to read this document

- **POL-xx** — Business Policies. How the mentorship programme operates. Change slowly, by stakeholder decision.
- **INV-xx** — Domain Invariants. Must be true at every commit boundary. No role, including Program Administrator, may violate one.
- **WFR-xx** — Workflow Rules. Which state transitions are legal, and under what guard.
- **AUTH-xx** — Authorization Rules. Who may perform an action, over which scope.

Each rule table carries an **Enforced at** column with one of:

Value	Meaning
DB	Database constraint, unique index, check constraint, or foreign key
TXN	Guard inside a database transaction, with row lock or version check
SVC	Domain service / application layer
MW	Request middleware (authentication, scope resolution)
EVT	Post-commit event handler

Rules enforced only at **SVC** and above are not enforced at all against direct database access or a future second write path. Where a rule is safeguarding-critical or data-integrity-critical, it is pushed down to **DB** or **TXN** deliberately.

2. Part 1 – Business Policies

2.1 Placement policies

ID	Policy	Enforced at	Notes
POL-001	A Placement concerns exactly one Student.	DB	One active Student Assignment per Placement. A second student is a second placement.
POL-002	A Placement belongs to exactly one Institution.	TXN	Derived , not independently editable — see POL-006.
POL-003	A Placement belongs to exactly one Facility.	TXN	Derived , not independently editable — see POL-006.
POL-004	A Placement always has exactly one active Ward Assignment. Ward changes close the current assignment and open a new one; history is never overwritten.	TXN	Rotation and mid-placement ward transfer are normal, not exceptional.
POL-005	A Placement has exactly one active Mentor Assignment at a time. Reassignment closes the current assignment and opens a new one.	TXN	Revised from draft — see POL-005a.
POL-006	<code>institution_id</code> and <code>facility_id</code> may be denormalised onto Placement for row-level scoping performance, but must always equal the value derived from Cohort (institution) and the active Ward Assignment (facility).	TXN	Derive, don't duplicate. They are a cached projection, never a second source of truth.
POL-007	The number of Placements in states <code>Orientation</code> , <code>Induction</code> , <code>Active</code> , or <code>Paused</code> in a Ward may not exceed that Ward's <code>max_students</code> .	TXN	Requires a row lock on the Ward during placement admission. See INV-16.
POL-008	A Cohort's <code>planned_capacity</code> is a planning target, not a constraint. Exceeding it warns; it never blocks.	SVC	Consistent with Stage 1 §9: capacity negotiation is deliberately out of the MVP as a workflow.
POL-020	A Placement carries both planned and actual dates: <code>planned_start</code> / <code>planned_end</code> (set at creation, never mutated) and <code>actual_start</code> / <code>actual_end</code> (set as the placement runs). A pause extends <code>actual_end</code> by the paused duration so the student's total mentorship time is preserved.	TXN	Confirmed by JKUAT, 30 July 2026 (OD-01 resolved). Both windows stay reportable, and interruption is measurable as the difference
POL-021	Every Pause Interval records a reason from a controlled vocabulary, so causes of training interruption are countable rather than buried in free text: <code>student_illness</code> , <code>student_personal</code> , <code>industrial_action</code> , <code>facility_closure</code> , <code>ward_capacity</code> , <code>academic_or_examinations</code> , <code>mentor_unavailability</code> , <code>safeguarding_investigation</code> , <code>public_health_emergency</code> , <code>other</code> (+ mandatory free text when <code>other</code>).	DB	Requested by Dr. Nyariki, 30 July 2026 — the analytical purpose of recording a pause is identifying <i>why</i> training is interrupted

POL-005a — Why co-mentorship is out of scope. The 1.0 draft allowed "additional mentors, with one marked Primary." That is dropped. Neither Stage 1, Stage 2, nor the concept note establishes co-mentorship, and its real shape is unknown: does a co-mentor have write access to sessions, or read only? Does the induction checklist need two sign-offs? Whose training completion gates the placement? Building a `is_primary` flag now means guessing at three unanswered questions and then defending the guess in the schema forever. Mentor Assignment already preserves history through open/close, so co-mentorship is a clean Phase 2 extension (`MentorAssignment.role`) rather than a Day-1 invariant.

2.2 Training and readiness policies

ID	Policy	Enforced at	Notes
POL-009	A Placement may not enter <code>Awaiting Student</code> unless the assigned Mentor has completed all modules marked <i>required</i> for the programme.	TXN	Controlled by programme flag <code>require_mentor_training_before_placement</code> (default: on). See OD-02.
POL-010	Training Module completion is facility-independent and outlives any single Placement. Reassigning or ending a placement never invalidates a completion record.	DB	Stage 1 §7 — Training Module is deliberately adjacent to, not owned by, Placement.
POL-011	Module completion requires a quiz score at or above the module's <code>pass_mark</code> . A failed attempt is recorded, not discarded.	SVC	Attempt history is evidence for CPD reporting.

2.3 Feedback and confidentiality policies

ID	Policy	Enforced at	Notes
POL-012	Submitted Feedback is append-only. A correction creates a new revision linked to the original; the original is never mutated or deleted.	DB	Immutability is the point — this is safeguarding-adjacent evidence.
POL-013	A Mentor may never read individual Feedback about themselves. Mentors see aggregates only, and only once a minimum of 3 submissions exist. The threshold applies to every figure actually displayed, not to the total — see POL-013a.	SVC	Confirmed by JKUAT, 30 July 2026 (OD-03 resolved). The min-n threshold is what stops "aggregate" from being a single identifiable student
POL-013a	No mentor-visible view may break an aggregate down by any student attribute — sex, age, year of study, course, registration cohort — or by any filter that narrows below the threshold. Suppression is evaluated per displayed cell, not per query.	SVC	A breakdown is a re-identification tool. Four responses filtered to "year 3" can be one student even though the total passed min-3. Raised by Dr. Nyariki, 30 July 2026
POL-013b	Sex and gender are not collected anywhere in the MVP. If sex-disaggregated reporting is later required for the research evaluation, the attribute is held on the University-owned Student master record, reported only at Cohort or Institution level under POL-013/013a, and never exposed in any mentor-facing view.	DB (field absent)	See OD-12

ID	Policy	Enforced at	Notes
POL-014	Feedback supports two modes. Confidential (default): attached to the Placement, readable by Nurse Manager, Coordinator, and Program Administrator, analysable per placement. Anonymous: attached to Cohort + Facility + Ward only — never to the Placement or Student.	DB	Both confirmed by JKUAT, 30 July 2026 (OD-06 resolved). A Placement identifies exactly one student, so anonymous feedback linked to a placement is not anonymous
POL-014a	Anonymity is chosen per submission by the student, defaulting to confidential. The interface must state, before submission, that an anonymous response cannot be followed up and will not appear in placement-level analysis.	SVC (UI contract)	Requested by Dr. Nyariki for students placed under a single mentor, where confidential still feels identifiable
POL-015	Every open-text field that can be submitted by any role must render the no-patient-identifiers instruction at the point of entry.	SVC (UI contract)	Carried from the Concept Note §8 data-protection rule.

2.4 Cross-institution data policies

ID	Policy	Enforced at	Notes
POL-016	The University side (Coordinator) may read Mentorship Session metadata — date, type, topics, duration — but not the free-text fields (<code>what_was_covered</code> , <code>feedback_given</code>).	SVC + MW	Stage 1 §8: hospital owns raw entries; university sees aggregate/summary unless escalated.
POL-017	The University side never reads Mentor master records (cadre, years of practice, digital literacy).	MW	Hospital staff records are hospital-owned.
POL-018	The Hospital side reads Student master records only for students placed with that facility.	MW	University owns the student master record.
POL-019	Escalation <i>counts and severities</i> may surface on any authorised dashboard. Escalation <i>description and action text</i> may not.	SVC	See INV-13.

3. Part 2 – Role Capability Summary

The full authorization model is Part 8. This section states the non-negotiables in plain language, because these are the sentences stakeholders will read.

A Student can view their own placement, complete orientation, submit feedback and confidence self-assessments, raise an escalation, and view their own session history. **A Student cannot** create a placement, assign a mentor, change a ward, approve their own induction, edit or delete a session, or see another student's anything.

A Mentor can complete training, complete the induction checklist for placements they are assigned to, log sessions for those placements, and review their own follow-ups. **A Mentor cannot** log or edit a session on a

placement they are not currently assigned to, edit another mentor's session, edit their own session after the correction window (POL/WFR-013), or read individual feedback about themselves.

A Nurse Manager can assign and reassign mentors within their facility, monitor induction and session activity, review and resolve escalations, pause and close placements, and export facility reports. **A Nurse Manager cannot** see another facility's placements, alter cohort membership, or act on a university-owned record.

A University Coordinator can create cohorts, enrol students, allocate students to facilities and wards, issue placement invitations, monitor their institution's placements across facilities, and export institution reports. **A University Coordinator cannot** access another institution's data, assign mentors, complete an induction, log a session, or read hospital staff records.

A Program Administrator can configure institutions, facilities, wards, users, modules, and content, and read programme-wide analytics. **A Program Administrator cannot** violate an invariant. Configuration authority is not domain authority — the Program Administrator has no path to log a session on someone's behalf, edit submitted feedback, or force an illegal state transition. (They can perform authorised *overrides* where a rule explicitly defines one, and every such override is audited with a mandatory reason.)

4. Part 3 – Domain Invariants

An invariant holds at every transaction boundary, for every actor, through every code path.

ID	Invariant	Enforced at	Error raised
INV-01	A Placement cannot exist without exactly one active Student Assignment.	DB	PlacementRequiresStudent
INV-02	A Placement cannot enter <code>Active</code> until its Induction Checklist is complete (all required items answered and signed by the mentor).	TXN	InductionIncomplete
INV-03	A Mentorship Session belongs to exactly one Placement — not to a Student, not to a Mentor.	DB	— (schema-level)
INV-04	A Mentorship Session's <code>session_date</code> must fall inside the Placement period, and must not fall inside a recorded Pause interval.	TXN	SessionOutsidePlacementWindow
INV-05	A Mentorship Session may only be created while the Placement is in state <code>Active</code> .	TXN	PlacementNotActive
INV-06	An Induction Checklist item may only be completed while the Placement is in state <code>Induction</code> .	TXN	InvalidTransition
INV-07	A Placement cannot transition to <code>Completed</code> while it has an Escalation in any state other than <code>Closed</code> , unless an authorised deferral override is recorded.	TXN	OpenEscalationBlocksCompletion
INV-08	A Placement cannot transition to <code>Completed</code> without a submitted final assessment (student exit survey + endline confidence rating).	TXN	FinalAssessmentMissing
INV-09	Placements in a terminal state (<code>Archived</code> , <code>Withdrawn</code>) are immutable. No field, no child record, no exception.	DB + TXN	PlacementArchived

ID	Invariant	Enforced at	Error raised
INV-10	No operational record is ever hard-deleted. Removal means soft-archive with the audit trail preserved.	DB	— (no delete endpoints exist)
INV-11	Every write is attributable to an authenticated actor, with actor, timestamp, action, target, and — where the action is an override — a reason.	TXN	UnattributedWrite
INV-12	A Placement has exactly one active Mentor Assignment and exactly one active Ward Assignment at any instant.	DB (partial unique index)	MentorAlreadyAssigned / WardAlreadyAssigned
INV-13	An Escalation's <code>description</code> and <code>action_taken</code> fields are readable only by the assigned reviewer, the submitter, and a Program Administrator. No dashboard, export, aggregate, or notification payload may contain them.	MW + SVC	EscalationAccessDenied
INV-14	Submitted Feedback and submitted Assessments are immutable. Corrections are new revisions.	DB	FeedbackImmutable
INV-15	All scored values stay inside their declared range: confidence and competency scores 0–100; Likert responses 1–5.	DB (check constraint)	ScoreOutOfRange
INV-16	Active placements in a Ward never exceed <code>Ward.max_students</code> .	TXN (row lock)	WardCapacityExceeded
INV-17	A Placement's denormalised <code>institution_id</code> / <code>facility_id</code> always equal the values derived from Cohort and the active Ward Assignment.	TXN	ScopeDerivationMismatch
INV-18	No field anywhere in the schema stores patient-identifiable data. Enforced structurally: no such field exists to write to.	DB (schema review gate)	—

On INV-07 and INV-08 together. The 1.0 draft contained a circularity: it said a placement reaches `Completed` when the final evaluation is submitted, *and* that a final assessment cannot exist before completion. Resolved here in one direction: the final assessment is a **precondition** of completion, not a consequence of it. It may be submitted from the placement's closing window onward (on or after the end date, or once closure is initiated), and INV-08 then gates the transition.

5. Part 4 – Workflow Rules

5.1 The Placement state model

Ten states. Eight linear, two lateral.

State	Meaning	Who is waiting on what
Draft	Coordinator is assembling the placement. Not visible to student or mentor.	Coordinator

State	Meaning	Who is waiting on what
Scheduled	Student, ward, and mentor are all assigned; dates are valid. Nothing has been communicated yet.	Coordinator, to invite
Awaiting Student	Placement invitation issued. Student has been told.	Student, to confirm
Orientation	Student confirmed; completing pre-arrival learning.	Student
Induction	Student has physically arrived; mentor is working through the labour ward checklist.	Mentor
Active	Normal mentorship. Sessions are logged, feedback submitted, escalations possible.	Mentor and Student
Paused	Temporarily suspended with a recorded reason (student illness, industrial action, facility closure).	Nurse Manager, to resume
Completed	Placement finished, final assessment submitted, escalations closed.	System, to archive
Archived	Read-only forever. Reportable.	Nobody — terminal
Withdrawn	Ended before completion. Read-only forever. Reason mandatory.	Nobody — terminal

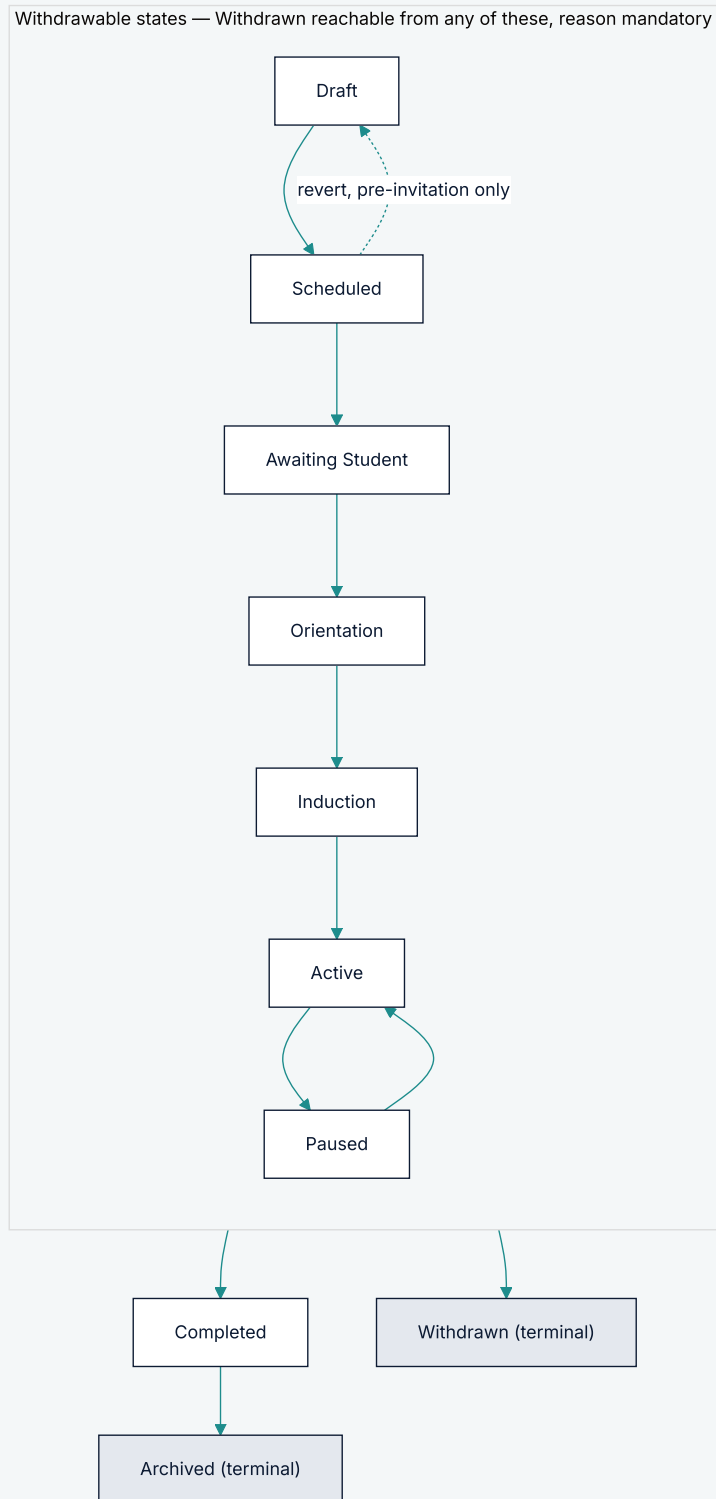
Follow-up Required is not a state. The Stage 2 draft floated it as an optional ninth state; it is formalised here as a **derived indicator on an Active placement**, computed from open follow-ups, missed session cadence, and open escalations. Making it a state would double the state space (every rule would need an **Active** -or- **Follow-up** clause) while adding no transition guard that isn't already expressible as a query.

5.2 The Placement transition table

This table is the enforced rule set. A transition not listed here does not exist, and attempting it raises `InvalidTransition`.

#	From	To	Guard (all must hold)	Initiated by
T-01	Draft	Scheduled	Student Assignment, Ward Assignment, and Mentor Assignment all exist; <code>start_date < end_date</code> ; ward capacity available (INV-16)	System-derived on assignment set becoming complete
T-02	Draft	Withdrawn	Reason recorded	Coordinator, Program Admin
T-03	Scheduled	Awaiting Student	Assigned mentor's required training complete (POL-009); placement invitation dispatched	Coordinator
T-04	Scheduled	Draft	—	Coordinator, Program Admin
T-05	Scheduled	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-06	Awaiting Student	Orientation	Student has confirmed the placement	System-derived on student confirmation

#	From	To	Guard (all must hold)	Initiated by
T-07	Awaiting Student	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-08	Orientation	Induction	Required orientation content complete or Nurse Manager override with recorded reason	Mentor, Nurse Manager
T-09	Orientation	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-10	Induction	Active	Induction Checklist complete and mentor-signed (INV-02)	System-derived on checklist completion
T-11	Induction	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-12	Active	Paused	Reason recorded; resume_expected_by optional	Nurse Manager, Coordinator, Program Admin
T-13	Active	Completed	End date reached or authorised early closure; final assessment submitted (INV-08); no non-Closed escalations, or deferral override recorded (INV-07)	Nurse Manager (Program Admin fallback)
T-14	Active	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-15	Paused	Active	Pause interval closed and recorded	Nurse Manager, Coordinator, Program Admin
T-16	Paused	Withdrawn	Reason recorded	Coordinator, Nurse Manager, Program Admin
T-17	Completed	Archived	Archive delay elapsed (programme-configurable, default 30 days) or explicit Program Admin action	System-derived (scheduled job) or Program Admin



Guards are omitted from the diagram; the transition table above is authoritative. **Active** → **Completed** requires the placement end date (or authorised early closure), a submitted final assessment, and no open escalations. **Induction** → **Active** requires a complete induction checklist.

WFR-001 — Why **Scheduled → **Draft** is allowed but **Awaiting Student** → **Draft** is not.** Before the invitation goes out, nothing has been communicated to anyone outside the coordinator's office; reverting to **Draft** because ward capacity fell through is bookkeeping. Once the student has been told they have a placement,

quietly returning it to planning erases a real-world communication. From `Awaiting Student` onward, backing out is `Withdrawn` — visible, reason-bearing, and auditable — not a silent disappearance.

WFR-002 — System-derived transitions. T-01, T-06, T-10, and T-17 are not buttons. They fire when their guard becomes satisfiable, and are audited against the actor whose action completed the guard. This matters for T-01 specifically: the assignment set is completed by *two different institutions* (Coordinator allocates the ward, Nurse Manager assigns the mentor), and neither has authority to press the other's button.

WFR-003 — Pause semantics. A Pause Interval is a first-class record — `started_at` (date of interruption), `ended_at` (date of resumption), `reason` from the POL-021 vocabulary, free text where required, and the acting actor. A placement may accumulate several. While paused: no sessions may be logged (INV-05), no induction items may be completed, and feedback and escalations remain open to the student (WFR-020).

On resumption, `actual_end` is extended by the paused duration (POL-020), preserving the student's total mentorship time while `planned_start` / `planned_end` retain the original window. Two figures therefore become directly reportable and are required indicators: **days interrupted** and **causes of interruption**, by facility and by cohort.

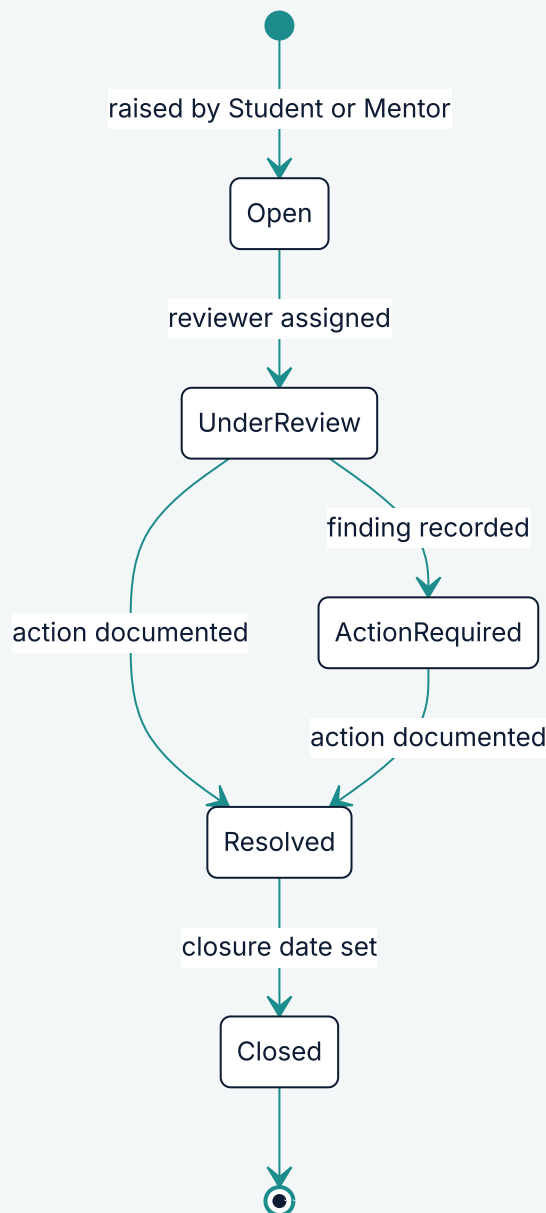
WFR-004 — Withdrawal semantics. `Withdrawn` is terminal and immutable, and requires a reason drawn from a controlled vocabulary (`student_withdrew` , `academic_decision` , `facility_capacity` , `safeguarding` , `administrative_error` , `other` + free text). A withdrawn placement remains fully reportable; it is not a delete.

5.3 The Escalation state model

Escalation has its own lifecycle, independent of Placement — established in Stage 2 and formalised here. Given that safeguarding is the highest-consequence workflow in the product, it does not get to be the one lifecycle without a transition table.

State	Meaning
<code>Open</code>	Raised. Notified. No reviewer yet.
<code>Under Review</code>	A reviewer is assigned and investigating.
<code>Action Required</code>	Investigation concluded that something must change; the action is not yet done.
<code>Resolved</code>	Action documented and complete.
<code>Closed</code>	Formally closed with a closure date. Terminal.

#	From	To	Guard	Initiated by
E-01	—	<code>Open</code>	Category, severity, and description present; no patient identifiers	Student, Mentor
E-02	<code>Open</code>	<code>Under Review</code>	Reviewer assigned, and reviewer is authorised for the escalation's scope	Nurse Manager, Coordinator, Program Admin
E-03	<code>Under Review</code>	<code>Action Required</code>	Finding recorded	Assigned reviewer
E-04	<code>Under Review</code>	<code>Resolved</code>	<code>action_taken</code> documented	Assigned reviewer
E-05	<code>Action Required</code>	<code>Resolved</code>	<code>action_taken</code> documented	Assigned reviewer
E-06	<code>Resolved</code>	<code>Closed</code>	<code>closure_date</code> set	Assigned reviewer, Nurse Manager, Program Admin



WFR-005 — No reopen. `Closed` is terminal. A recurrence is a **new** Escalation, optionally carrying `related_escalation_id`. Reopening would make the audit trail of a safeguarding record ambiguous about when it was and was not under active review.

WFR-006 — External referral is an outcome, not a state. The SADD's "escalated further" (to institutional safeguarding processes outside the system) is recorded as a resolution outcome (`resolution_outcome = referred_externally`) with a referral note, not as a sixth state. The system cannot model a workflow it does not own.

WFR-007 — Severity-based routing. On `Open` : notify the facility's Nurse Manager always; additionally notify the University Coordinator when `severity = high` . This routes notification only — never content (INV-13).

WFR-008 — Reviewer conflict of interest. The subject of an escalation may not be its reviewer. Where the only facility-scoped reviewer is the subject, assignment escalates to the University Coordinator or Program Administrator.

6. Part 5 – Mentorship Session Rules

ID	Rule	Enforced at	Error
WFR-009	A Session requires: Placement, Mentor, <code>session_date</code> , <code>session_type</code> , and at least one topic.	DB	<code>SessionValidationFailed</code>
WFR-010	Optional: <code>what_was_covered</code> , <code>feedback_given</code> , <code>follow_up_action</code> , <code>time_spent_min</code> .	—	—
WFR-011	The Mentor on a Session must be the Placement's active Mentor Assignment at the time of creation.	TXN	<code>UnauthorizedFacilityAccess</code>
WFR-012	A Mentor may never create, edit, or void a Session belonging to another Mentor.	MW + TXN	<code>UnauthorizedFacilityAccess</code>
WFR-013	A Mentor may correct their own Session within 24 hours of creation. After that it is append-only: a correction is a linked revision.	TXN	<code>SessionCorrectionWindowClosed</code>
WFR-014	Duplicate rule: a Session is a duplicate if it matches an existing Session on Placement + Mentor + <code>session_date</code> and was submitted within a 5-minute window of it. Duplicates are rejected idempotently — the client receives the original record, not an error page.	TXN (+ client idempotency key)	<code>DuplicateSession</code>
WFR-015	A Session may not be logged while the Placement is in any state other than <code>Active</code> (INV-05), and its <code>session_date</code> may not fall inside a Pause interval (INV-04).	TXN	<code>PlacementNotActive</code> / <code>SessionOutsidePlacementWindow</code>

On WFR-014. The draft rule was "same mentor, same placement, same time." Exact timestamp equality essentially never occurs — even an accidental double-tap is tens or hundreds of milliseconds apart, and an offline draft replayed twice can be minutes apart. The rule needs a real window and an idempotency key from the client, because the offline-first design in the Concept Note guarantees retry-driven double submission will happen in the field.

7. Part 6 – Assessment & Feedback Rules

ID	Rule	Enforced at	Error
WFR-016	Feedback and Assessments are immutable once submitted; corrections are new revisions (INV-14).	DB	<code>FeedbackImmutable</code>
WFR-017	Confidence and competency scores are 0–100; Likert responses are 1–5 (INV-15).	DB	<code>ScoreOutOfRange</code>
WFR-018	A baseline confidence self-assessment may only be submitted while the Placement is in <code>Orientation</code> or <code>Induction</code> .	SVC	<code>AssessmentOutOfWindow</code>

ID	Rule	Enforced at	Error
WFR-019	A final assessment (exit survey + endline confidence) may only be submitted on or after the Placement's <code>end_date</code> , or once closure has been initiated. It is a precondition of <code>Completed</code> (INV-08).	TXN	<code>AssessmentOutOfWindow</code>
WFR-020	Periodic feedback may be submitted at any time while the Placement is in <code>Orientation</code> , <code>Induction</code> , <code>Active</code> , or <code>Paused</code> . A paused student must keep their voice.	SVC	<code>PlacementNotActive</code>
WFR-021	Confidence change is derived (endline – baseline), never stored as an editable field.	SVC	—
WFR-022	Progress Summary (training %, induction %, session cadence, assessment %) is computed on read, never authored.	SVC	—

8. Part 7 – Notification Rules

ID	Rule	Enforced at
WFR-023	The Notification context never decides. It executes. It contains no business logic and reads no domain state to make a decision.	EVT
WFR-024	Notifications are triggered exclusively by domain events, never by direct calls from another context.	EVT
WFR-025	A notification payload may never contain an Escalation <code>description</code> or <code>action_taken</code> , individual Feedback text, or any free-text session field. Payloads carry identifiers and a template key.	EVT
WFR-026	Notification dispatch failure never fails the originating business transaction. Events are written to a transactional outbox in-transaction and dispatched after commit, with retry.	TXN + EVT

WFR-026 is a correction to how "atomicity" was described in the 1.0 draft — see §10.

9. Part 8 – Authorization Rules

9.1 The scope model

Every authenticated request resolves to a **scope tuple** at the middleware layer, derived from the user record — never from client-supplied parameters:

```
scope = { program_id, role, institution_id?, facility_id?, ward_ids?, user_id }
```

ID	Rule	Enforced at
AUTH-001	Scope is derived server-side from the authenticated user on every request. A scope value present in a request body or query string is ignored.	MW
AUTH-002	Every query against a scoped entity is filtered by scope at the data-access layer, not the view layer. An endpoint that forgets its filter must return nothing, not everything.	MW + SVC
AUTH-003	Every user belongs to exactly one organisation — an Institution or a Facility — except Program Administrators, who are scoped to a Programme.	DB
AUTH-004	Cross-scope reads are impossible by construction, not by UI omission. Direct API access with a valid token from another facility returns 404, not 403 (a 403 confirms the record exists).	MW
AUTH-005	Field-level restrictions (INV-13, POL-016, POL-017) are applied by the serialisation layer, so no code path can leak a restricted field by returning a "full" object.	SVC

9.2 Action matrix

✓ permitted · A aggregate/derived only · M metadata only, no free text · — denied

Action	Student	Mentor	Nurse Manager	Coordinator	Program Admin
Create institution / facility / ward	—	—	—	—	✓
Create cohort, enrol students	—	—	—	✓ (own institution)	✓
Create placement, allocate ward	—	—	—	✓ (own institution)	✓
Assign / reassign mentor	—	—	✓ (own facility)	—	✓
Issue placement invitation	—	—	—	✓	✓
Confirm placement	✓ (own)	—	—	—	—
Complete induction checklist	—	✓ (assigned)	✓ (own facility)	—	—
Log mentorship session	—	✓ (assigned)	—	—	—
Read session free text	✓ (own)	✓ (own entries)	✓ (own facility)	M	✓
Submit feedback / confidence	✓ (own)	—	—	—	—
Read individual feedback	✓ (own)	A (min 3)	✓ (own facility)	✓ (own institution)	✓
Raise escalation	✓	✓	✓	✓	✓
Read escalation description	✓ (own submissions)	✓ (own submissions)	✓ (if assigned reviewer)	✓ (if assigned reviewer)	✓

Action	Student	Mentor	Nurse Manager	Coordinator	Program Admin
Read escalation counts / severity	—	—	✓ (own facility)	✓ (own institution)	✓
Review / resolve / close escalation	—	—	✓ (own facility)	✓ (high severity, own institution)	✓
Pause / resume placement	—	—	✓	✓	✓
Complete placement	—	—	✓	—	✓
Withdraw placement	—	—	✓	✓	✓
Archive placement	—	—	—	—	✓
Complete training module	—	✓ (own)	✓ (own)	—	—
Publish / edit training content	—	—	—	—	✓
Read mentor master record	—	✓ (own)	✓ (own facility)	—	✓
Read student master record	✓ (own)	✓ (assigned)	✓ (placed at facility)	✓ (own institution)	✓
Export facility report	—	—	✓ (own facility)	—	✓
Export institution report	—	—	—	✓ (own institution)	✓
Export programme report	—	—	—	—	✓

AUTH-006 — Nurse Manager scope is the facility, filterable by ward. At pilot scale a facility is one maternity unit, so facility-scoped and ward-scoped are the same set; ward-level narrowing exists as a filter, not as an authorization boundary. Revisit if a pilot facility runs more than one mentoring unit — see **OD-04**.

AUTH-007 — Coordinator escalation authority is severity-gated. A Coordinator may review escalations from their own institution's placements at `severity = high`, matching Stage 1 §8 ("University sees it only above a severity threshold or on resolution failure"). They are not a general facility reviewer.

10. Part 9 — Transaction Boundaries

Each row is one atomic unit. Everything inside commits together or not at all. Everything after the arrow happens **after** commit, driven by the outbox.

ID	Operation	Inside the transaction	After commit (outbox)
TX-01	Create placement	Placement row; Student Assignment; Ward Assignment; ward capacity check with row lock; derived scope fields; audit entry; <code>PlacementCreated</code> event row	Notifications; dashboard projection refresh
TX-02	Assign / reassign mentor	Close current Mentor Assignment; open new one; partial-unique check (INV-12); T-01 guard re-evaluation; audit entry; <code>MentorAssigned</code> event row	Notify mentor and student; dashboards
TX-03	Complete induction	Final checklist item; checklist status; T-10 transition to <code>Active</code> ; audit entries; <code>InductionCompleted</code> + <code>PlacementActivated</code> event rows	Notifications; progress recomputation
TX-04	Log session	Duplicate check (WFR-014); state and window guards (INV-04, INV-05); Session row; audit entry; <code>SessionLogged</code> event row	Progress recomputation; follow-up flag; supervisor notification if <code>follow_up = supervisor_review</code>
TX-05	Submit feedback	Feedback row (immutable); anonymity routing (POL-014); audit entry; <code>FeedbackSubmitted</code> event row	Analytics projection
TX-06	Raise escalation	Escalation row in <code>Open</code> ; audit entry; <code>EscalationRaised</code> event row	Severity-based notification routing (WFR-007)
TX-07	Resolve / close escalation	Guard check (E-04/E-05/E-06); status; <code>action_taken</code> ; <code>closure_date</code> ; audit entry; <code>EscalationClosed</code> event row	Notify submitter (status only, never content); dashboards
TX-08	Complete placement	INV-07 and INV-08 guards; state change; audit entry; <code>PlacementCompleted</code> event row	Final report generation; certificate; notifications
TX-09	Export data	Export record with actor, scope, filters, row count; audit entry	File generation and delivery

ID	Rule	Enforced at
WFR-027	No external I/O — email, SMS, file generation, third-party API — occurs inside a business transaction.	SVC
WFR-028	Every state transition takes a row lock (or optimistic version check) on the Placement, so two concurrent transitions cannot both pass their guard.	TXN
WFR-029	Ward capacity checks (INV-16) take a lock on the Ward row, not a read-then-write.	TXN

On "all or nothing". The 1.0 draft listed notification and dashboard updates inside the atomic unit for mentor assignment. That is not implementable: an email provider timeout would roll back a legitimate assignment, and a committed email cannot be un-sent by a rollback. The correct boundary is the transactional outbox above — the *decision* is atomic, the *side effects* are eventually consistent and retryable.

11. Part 10 – Audit Rules

ID	Rule	Enforced at
WFR-030	Every write produces an append-only Audit entry: actor, role, scope, timestamp, action, entity type, entity id, before/after for changed fields.	TXN
WFR-031	Audit entries are immutable and never deleted, including when their subject record is soft-archived.	DB
WFR-032	Reads are audited for three categories only: Escalation description access, individual Feedback access, and data exports. Auditing all reads at pilot scale buys noise, not safety.	SVC
WFR-033	Every override permitted by a rule in this document (T-08 orientation override, INV-07 deferral, POL-009 training bypass) requires a mandatory free-text reason and is audited as an override, not as a normal write.	TXN
WFR-034	Audit entries contain no restricted field values — an audit of an escalation edit records that the description changed, not what it changed to.	SVC

12. Part 11 – Domain Error Vocabulary

Shared verbatim across backend, frontend, and test suites. One error code, one meaning, one message catalogue.

Code	HTTP	Meaning
PlacementRequiresStudent	422	Placement write would leave it without an active student assignment
PlacementAlreadyActive	409	Transition attempted on an already-active placement
PlacementNotActive	409	Action requires state <code>Active</code> (session logging)
PlacementClosed	409	Placement is <code>Completed</code> ; operational writes rejected
PlacementArchived	409	Placement is in a terminal state and immutable
InvalidTransition	409	Requested state transition is not in the transition table
InductionIncomplete	409	Activation blocked; required checklist items outstanding
FinalAssessmentMissing	409	Completion blocked; exit survey / endline confidence not submitted
OpenEscalationBlocksCompletion	409	Completion blocked by a non- <code>Closed</code> escalation
MentorAlreadyAssigned	409	An active mentor assignment already exists
WardAlreadyAssigned	409	An active ward assignment already exists
StudentAlreadyAssigned	409	Student already has an overlapping active placement
WardCapacityExceeded	409	Ward <code>max_students</code> would be exceeded
TrainingIncomplete	409	Assigned mentor has not completed required modules
DuplicateSession	200	Idempotent duplicate; original record returned

Code	HTTP	Meaning
SessionOutsidePlacementWindow	422	session_date outside placement period or inside a pause interval
SessionValidationFailed	422	Required session fields missing
SessionCorrectionWindowClosed	409	Session edit attempted after the 24-hour window
FeedbackImmutable	409	Edit attempted on submitted feedback or assessment
AssessmentOutOfWindow	409	Assessment submitted outside its permitted placement states
ScoreOutOfRange	422	Score outside its declared range
InvalidEscalationTransition	409	Escalation transition not in the escalation table
EscalationAccessDenied	404	Restricted escalation field requested by an unauthorised actor
ReviewerConflictOfInterest	409	Assigned reviewer is the subject of the escalation
UnauthorizedFacilityAccess	404	Cross-scope access attempt (404, never 403 — see AUTH-004)
ScopeDerivationMismatch	500	Denormalised scope fields diverged from derived values — an integrity bug, not user error
UnattributedWrite	500	Write attempted with no authenticated actor — an integrity bug
OverrideReasonRequired	422	Override attempted without a reason

13. Part 12 — Open Decisions Register

These are product decisions, not engineering ones. Each is logged rather than guessed, with a recommendation and the consequence of deferring. This is the same discipline applied to the capacity-negotiation question in Stage 1 §9.

ID	Question	Status	Resolution
OD-01	When a Placement is paused, does the end date extend, or does it resume inside the original window with less active time?	Resolved — JKUAT, 30 July 2026	Extend. actual_end moves by the paused duration; planned_start / planned_end retain the original window (POL-020). Confirmed implicitly by the request to record date of interruption, date of resumption, and date of completion separately. Stage 5 unblocked
OD-02	Should mentor training completion hard-block a placement reaching Awaiting Student ?	Decided by Sinaps	Default on , per-programme flag, Program Admin override with a mandatory reason (WFR-033)
OD-03	Feedback visibility model and the min-n aggregate threshold	Resolved — JKUAT, 30 July 2026	Nurse Manager and Coordinator read individual responses; Mentor sees aggregates at $n \geq 3$. Threshold applies per displayed figure , and no mentor-visible breakdown by any student attribute (POL-013, POL-013a)
OD-04	Is a Nurse Manager's authorization boundary the facility or the ward?	Decided by Sinaps	Facility, ward as filter (AUTH-006)

ID	Question	Status	Resolution
OD-05	Archive delay after completion	Decided by Sinaps	30 days, programme-configurable
OD-06	What does anonymous feedback attach to?	Resolved — JKUAT, 30 July 2026	Both modes supported. Confidential is the default and links to the Placement; anonymous attaches to Cohort + Facility + Ward only, chosen per submission, with the trade-off stated in the interface before submission (POL-014, POL-014a)
OD-07	Who may authorise early completion before the end date?	Decided by Sinaps	Nurse Manager with a reason; Coordinator notified, not consulted
OD-08	Can a Placement complete with a low-severity escalation still open (INV-07)?	Decided by Sinaps	Yes, via explicit deferral override by Coordinator or Program Admin with a reason. High severity is never deferrable
OD-12	Does the research evaluation require sex-disaggregated reporting?	Open — asked of JKUAT, 30 July 2026	If yes: attribute lives on the University-owned Student master record, reported at Cohort/Institution level only under POL-013/013a, never in a mentor-facing view (POL-013b). If no: the field is not collected at all. Blocks nothing; affects Stage 5 only if the answer is yes

14. Part 13 – The MentorMAMA Constitution

One page. Every contributor reads this before writing code.

Identity Every user belongs to exactly one organisation, or is a Programme Administrator. Every action is attributable to an authenticated actor. Scope is derived on the server; the client is never trusted with it.

Placement A placement concerns one student, at one facility, in one ward at a time, with one mentor at a time. It owns every operational record produced during it: induction, sessions, progress, assessments. Its `institution_id` and `facility_id` are derived values, cached for speed, never authored. It moves only along the transition table in §5.2, and it cannot become `Active` until induction is complete.

Mentorship Sessions belong to placements, not to people. They may only be logged while the placement is `Active`, only by the currently assigned mentor, only within the placement window, and never inside a pause. A mentor never touches another mentor's records.

Assessment Feedback and assessments are immutable once submitted; corrections are revisions. Progress is computed, never stored as an editable truth. A mentor never sees individual feedback about themselves.

Safeguarding Escalation content is readable only by the assigned reviewer, the submitter, and a Programme Administrator — enforced in serialisation, not in the UI. Escalation has its own lifecycle and never redefines a placement's. Counts travel to dashboards; content never does. The subject of an escalation never reviews it.

Audit No operational data is ever hard-deleted. Every write is logged. Every override carries a reason. Terminal states are immutable, permanently.

Boundaries Notification executes; it never decides. Contexts communicate through domain events, not direct calls. Decisions are atomic; side effects are eventually consistent and retryable. No external I/O inside a transaction.

Privacy No patient-identifiable data exists anywhere in the schema. There is no field to write it to.

15. What Stage 3.5 Delivers Next

Everything to this point is static: entities, rules, relationships. Stage 3.5 makes the system **move**, before any module boundary or endpoint is drawn. For each of these workflows we will produce a sequence diagram and a step table naming the actor, the guards checked in order, the services involved, the events raised, the records written, and the transaction boundary:

1. Student onboarding and placement invitation
2. Placement creation and ward allocation
3. Mentor assignment and reassignment
4. Orientation → arrival → induction completion → activation
5. Mentorship session logging (including the offline retry path)
6. Feedback and assessment submission (including anonymous routing)
7. Escalation raise → review → resolution → closure
8. Placement completion, reporting, and archival

This is the artifact that makes Stage 4 (module architecture) and the API surface obvious rather than invented. It is also where the original SADD's weakest area gets repaired: it specified REST endpoints (§6.3) without ever having walked through what actually happens, in order, when a session is logged.

Appendix A – Traceability

Source	Where it lands in Stage 3
Stage 1 §7 (Placement as aggregate root)	POL-001-007, INV-01-12
Stage 1 §8 (Data ownership table)	POL-016-019, INV-13, AUTH matrix
Stage 1 §9 (Capacity decision)	POL-008
Stage 2 (Placement lifecycle, corrected)	§5.1, §5.2 — including Paused and Withdrawn
Stage 2 §9.2 (Escalation independence)	§5.3, INV-07
Stage 2 (Domain events)	WFR-023-026, TX-01-09
Concept Note §8 (Data capture / no patient data)	POL-015, INV-18
Concept Note §13 (Privacy & ethics)	INV-13, WFR-030-034, AUTH-001-005
Concept Note Appendix A (Session log)	WFR-009-015
Concept Note Appendix C (Permission matrix)	§9.2
SADD §3.3 (Escalation lifecycle)	§5.3, WFR-006, WFR-007
SADD §6.2 (Multi-tenancy scoping)	POL-006, INV-17, AUTH-001-004
SADD §5.5 (Auditability)	WFR-030-034

Appendix B – Revision Notes

B.1 – v1.2: JKUAT decisions of 30 July 2026

#	Change	Source
1	POL-020 added: planned and actual dates both stored; a pause extends <code>actual_end</code> . OD-01 resolved, Stage 5 unblocked	Dr. Nyariki's request to record date of interruption, resumption, and completion separately
2	POL-021 added: pause reason drawn from a controlled ten-value vocabulary rather than free text, so causes of training interruption are countable. WFR-003 rewritten accordingly, and "days interrupted" plus "causes of interruption" become required indicators	Dr. Nyariki — the analytical purpose of recording a pause is identifying why training is interrupted
3	POL-013a added: the min-3 threshold applies to every displayed figure , not the total, and no mentor-visible view may break an aggregate down by any student attribute	Dr. Nyariki's question about sex/gender being re-identifying. The mechanism generalises past gender: any breakdown is a re-identification tool
4	POL-013b added: sex and gender are not collected in the MVP; if later required, they live on the University-owned student record and never reach a mentor-facing view	Same
5	POL-014 rewritten: both confidential (default, placement-linked) and anonymous (cohort/facility/ward only) modes are supported, chosen per submission (POL-014a), with the trade-off stated before submission. OD-03 and OD-06 resolved	Dr. Nyariki approved confidential as default and asked that anonymity remain possible where a student is placed under a single mentor
6	OD-12 opened: does the research evaluation need sex-disaggregated reporting?	Follow-up question arising from item 3

B.2 – v1.1: corrections to the v1.0 draft

#	Change	Reason
1	Transition table extended from 8 linear states to the full 10, adding <code>Paused</code> and <code>Withdrawn</code> rows	The v1.0 table silently dropped the lateral states that Stage 2's correction had introduced. Since the table is the enforced rule set, shipping it as-is would have deleted that correction from the implementation.
2	<code>Scheduled</code> → <code>Draft</code> resolved from "maybe, depends" to: permitted, but not from <code>Awaiting Student</code> onward (WFR-001)	A business rule cannot ship as "depends."
3	Co-mentorship (<code>is_primary</code>) removed; POL-005 simplified to one active Mentor Assignment	Scope creep by the same test applied to capacity negotiation in Stage 1 — it solves a problem the pilot does not have, with three unanswered sub-questions.
4	Added INV-05 (sessions only while <code>Active</code>) and INV-07 (no completion with an open escalation)	Both were implied by Stage 2 but never written down. Without INV-05, logging a session on a <code>Draft</code> placement was forbidden by nothing.
5	Escalation given its own state model, transition table, and access invariant (§5.3, INV-13)	v1.0 formalised Placement transitions in detail and left the safeguarding-critical lifecycle with no formal rules at all.

#	Change	Reason
6	POL-002/003 rewritten as derived-not-duplicated, with INV-17 to enforce it	The SADD wants <code>facility_id / institution_id</code> denormalised for row-level scoping; without an invariant they become two independently-editable fields that usually agree.
7	Duplicate-session rule given a 5-minute window and an idempotency key (WFR-014)	Exact timestamp equality never occurs; offline retry guarantees duplicates minutes apart.
8	Final-assessment circularity resolved: it is a precondition of <code>Completed</code> , not a consequence (INV-08, WFR-019)	v1.0 stated both directions and was unimplementable as written.
9	Notification/dashboard side effects moved out of the atomic unit into a transactional outbox (WFR-026, §10)	v1.0's "everything or nothing" would have rolled back valid assignments on an email timeout, and cannot un-send a sent email.
10	Added INV-16 / POL-007 (ward capacity) and WFR-028/029 (locking)	<code>Ward.max_students</code> exists in the Concept Note data model but was never enforced by any rule.
11	Added POL-013 min-n threshold and POL-014 anonymous-routing rule	"Mentor sees aggregate" re-identifies the student when the aggregate is one response; anonymous feedback attached to a placement is not anonymous.
12	Follow-up Required formalised as a derived indicator, not a state (§5.1)	Avoids doubling the state space for a condition already expressible as a query.
13	Added WFR-008 (reviewer conflict of interest) and AUTH-007 (severity-gated coordinator authority)	Both follow directly from Stage 1 §8 but had no rule.
14	Open Decisions Register added (§13), including the pause/end-date question	Open questions get logged, with a recommendation and a blocking assessment — not left implicit.

End of Stage 3: Business Rules & Domain Invariants — Sinaps Technology / JHUB Africa. Confidential.